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Math 321

Lab 6: Comparing Means and Linear Regression

Your job: Complete Exercises 1 thru 5. This should be written in a Word document, including all appropriate explanations and labeled graphs. You are required to work with a partner. Please save your final copy as a PDF

Exercise 1: The examples above showed how to work with ANOVA for length vs. years. What about weight? Is there a difference in average **weight** for fish, depending on what year they were caught? Use ANOVA to conduct a complete hypothesis (with $\alpha = 0.05$) and include the following:

- null and alternative hypotheses,
- side by side boxplots for the different years,
- MSG, MSE, F statistic, and p-value,
- conclusions (related to the context, and understandable to the general public)
- If you reject the null hypothesis, follow-up with the TukeyHSD test and clearly indicate which years have statistically significant differences in average weight.

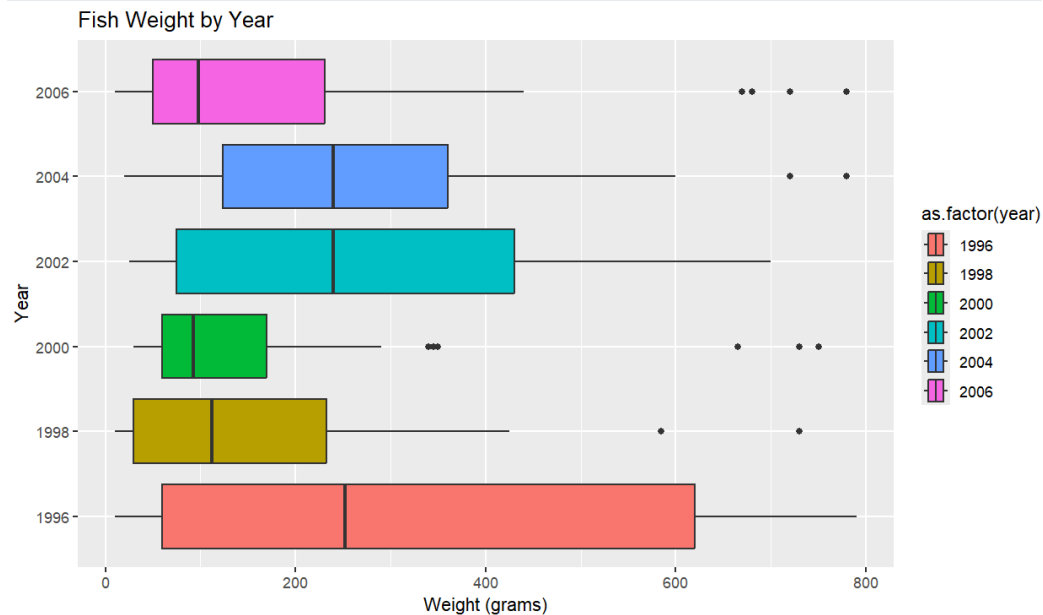
Null Hypothesis (H_0):

The mean fish weight is the same for all years:

$$\mu_{1996} = \mu_{1998} = \mu_{2000} = \mu_{2002} = \mu_{2004} = \mu_{2006}$$

Alternative Hypothesis (H_1):

At least one year has a different mean fish weight



```
> summary(fish_weight_ANOVA)
              Df Sum Sq Mean Sq F value    Pr(>F)
as.factor(year)  5  669954   133991     3.208 0.00833 **
Residuals      194  8103992    41773
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

MSG =133991 | MSE =41773 | F =3.208 | Pvalue = 0.00833

Conclusion: We reject the null hypothesis. This indicates that there is a statistically significant difference in the average fish weight across the years the fish were caught. To find out which years differ, a Tukey HSD test should be performed. This follow-up test will identify the specific years where average fish weights are significantly different from each other. Fish weight has not remained consistent over the years some years had heavier or lighter fish on average.

```
> TukeyHSD(fish_weight_ANOVA)
Tukey multiple comparisons of means
 95% family-wise confidence level

Fit: aov(formula = weight ~ as.factor(year), data = fish)

$`as.factor(year)`
              diff       lwr       upr     p adj
1998-1996 -175.9834559 -352.54718   0.5802728 0.0513025
2000-1996 -174.1685520 -338.53051  -9.8065910 0.0308218
2002-1996 -72.5054466 -254.65281  109.6419201 0.8615036
2004-1996 -72.7647059 -251.35866  105.8292440 0.8493160
2006-1996 -155.8123249 -324.92642   13.3017684 0.0900035
2000-1998   1.8149038 -130.36514  133.9949453 1.0000000
2002-1998 103.4780093  -50.25674  257.2127558 0.3826268
2004-1998 103.2187500  -46.28880  252.7263017 0.3534166
2006-1998  20.1711310 -117.87338  158.2156370 0.9983003
2002-2000 101.6631054  -37.88788  241.2140863 0.2932313
2004-2000 101.4038462  -33.47614  236.2838372 0.2595488
2006-2000  18.3562271 -103.69454  140.4069936 0.9980480
2004-2002  -0.2592593 -156.32149  155.8029749 1.0000000
2006-2002 -83.3068783 -228.42476   61.8109998 0.5649281
2006-2004 -83.0476190 -223.67953   57.5842916 0.5335900
```

Exercise 2: Is there a difference in average length for fish, depending on species? Use ANOVA to conduct a complete hypothesis (with $\alpha = 0.05$) and include the following:

- null and alternative hypotheses,
- side by side boxplots for the different species,
- MSG, MSE, F statistic, and p-value, and
- conclusions (related to the context, and understandable to the general public)
- If you reject the null hypothesis, follow-up with the `pairwise.t.test()` with `p.adj="bonf"` and clearly indicate which species have statistically significant differences in average length. It may be convenient to copy the results from the `pairwise.t.test()` into your lab and highlight the ones that indicate statistically significant differences.

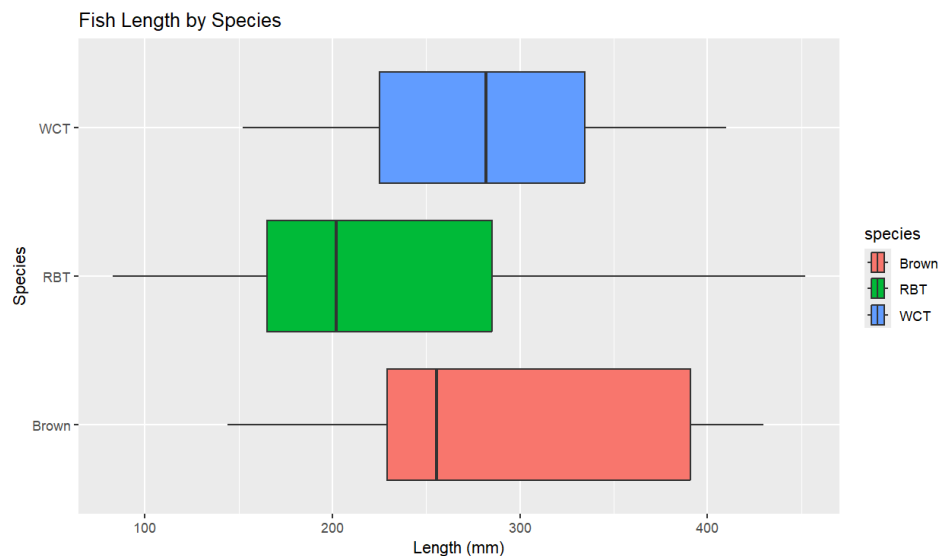
Null Hypothesis (H_0):

The mean fish length is the same for all species:

$$\mu_{\text{species1}} = \mu_{\text{species2}} = \mu_{\text{species3}} = \dots$$

Alternative Hypothesis (H_1):

At least one species has a different average fish length



```
> summary(fish_species_ANOVA)
      Df Sum Sq Mean Sq F value    Pr(>F)
species    2  170165    85082   11.72 1.55e-05 ***
Residuals 197 1429940     7259
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

MSG =85082| MSE =7259 | F =11.72 | Pvalue = 0.0000155

Conclusion: We reject the null hypothesis at the $\alpha = 0.05$. This means there is a statistically significant difference in average fish length between at least some of the species. The small p-value indicates that the differences observed are very unlikely to be due to random chance alone.

To determine which species, differ, we conducted a Bonferroni-corrected pairwise t-test. This test identified specific pairs of species whose average lengths are significantly different from one another, providing clearer insight into which species vary in size. Fish length depends on species, some species are, on average, longer or shorter than others.

```
> pairwise.t.test(fish$length, fish$species, p.adj = "bonf")

Pairwise comparisons using t tests with pooled SD

data:  fish$length and fish$species

      Brown   RBT
RBT 0.00022 -
WCT 1.00000 0.00162

P value adjustment method: bonferroni
```

- RBT differs significantly in average length from both Brown and WCT species.
Brown and WCT do not differ significantly from each other in average length.

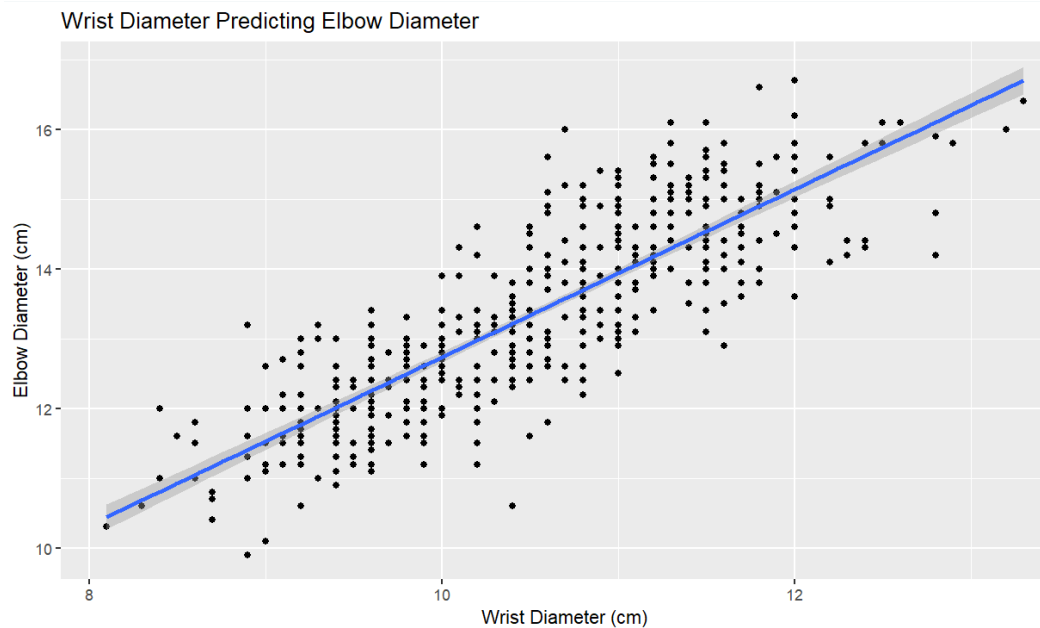
Exercise 3: For each of the following, create scatterplots, provide R^2 values, and provide the equation of the regression lines for:

- Wrist diameter (wri_di) predicting elbow diameter (elb_di)
- Knee diameter (kne_di) predicting elbow diameter (elb_di)
- Height (hgt) predicting waist circumference (wai_gi)

Then respond to the following:

- Which of the plots shows the strongest relationship? Describe how you know from the plot and from the summary statistics.
- Which of the plots shows the weakest relationship? Describe how you know from the plot and from the summary statistics. Even though this is the weakest relationship, is it still statistically significant?

- For the height vs waist circumference problem, what is the real-world meaning of the slope of your regression line?

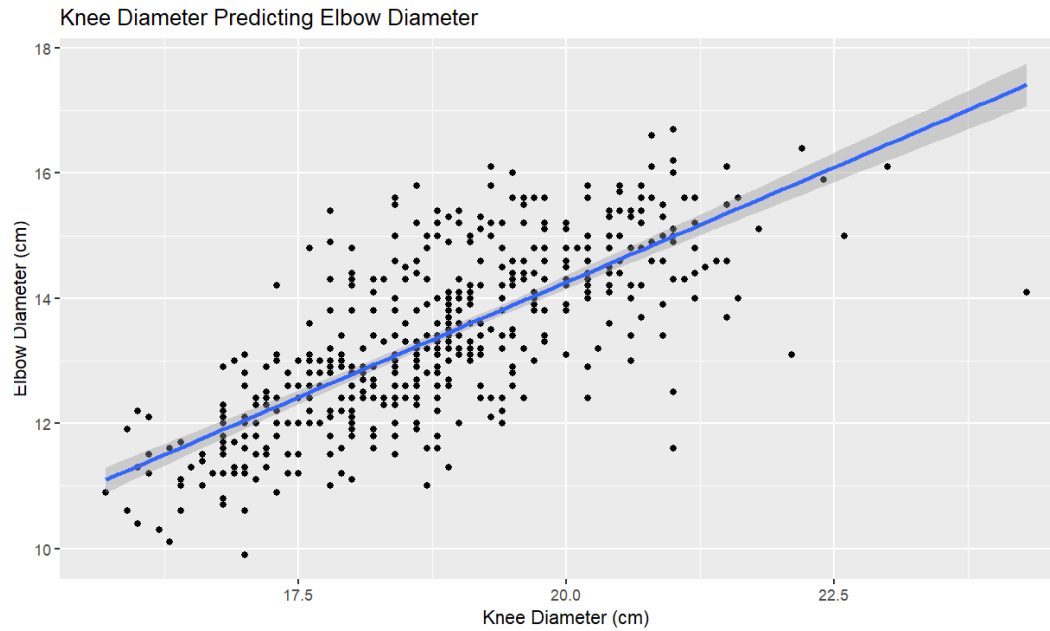


```
Call:
lm(formula = elb_di ~ wri_di, data = bdims)

Residuals:
    Min       1Q   Median       3Q      Max
-2.61361 -0.51032  0.02375  0.46111  2.42540

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.6993     0.3662    1.91  0.0567 .
wri_di        1.2033     0.0346   34.78 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7349 on 505 degrees of freedom
Multiple R-squared:  0.7055,    Adjusted R-squared:  0.7049
F-statistic: 1210 on 1 and 505 DF,  p-value: < 2.2e-16
```

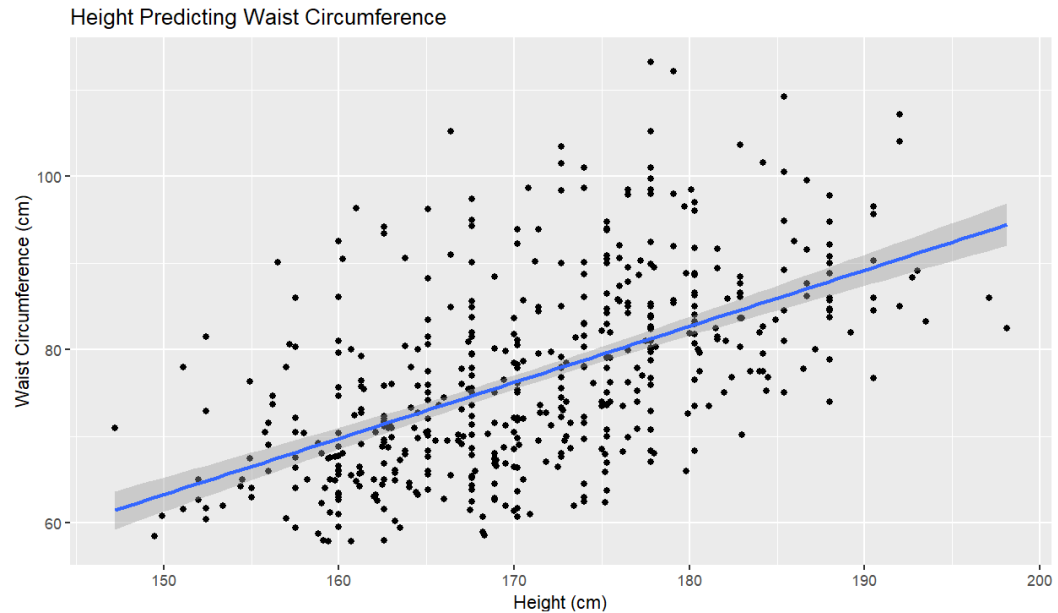


```
Call:
lm(formula = elb_di ~ kne_di, data = bdims)

Residuals:
    Min       1Q   Median       3Q      Max
-3.3930 -0.5719 -0.0227  0.5171  2.7570

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.42909    0.57446  -0.747   0.455
kne_di       0.73439    0.03046  24.109 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9234 on 505 degrees of freedom
Multiple R-squared:  0.5351,    Adjusted R-squared:  0.5342
F-statistic: 581.3 on 1 and 505 DF,  p-value: < 2.2e-16
```



```
Call:
lm(formula = wai_gi ~ hgt, data = bdim)

Residuals:
    Min       1Q   Median       3Q      Max
-17.205  -6.507  -1.288   4.721  31.912

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -33.80729    7.43970  -4.544 6.91e-06 ***
hgt          0.64733    0.04341  14.914 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.185 on 505 degrees of freedom
Multiple R-squared:  0.3058,    Adjusted R-squared:  0.3044 
F-statistic: 222.4 on 1 and 505 DF,  p-value: < 2.2e-16
```

Strongest Relationship:

- **Wrist _Elbow (0.7055)**, showing the strongest linear relationship.
- The graph shows points tightly clustered around the regression line. Each 1 cm increase in wrist diameter +1.20 cm elbow diameter

Weakest Relationship:

- **Height _Waist (0.3058)**.
- The points are more widespread, suggesting a weaker relationship. Each 1 cm increase in height +0.65 cm waist circumference

Despite being the weakest, the p-value for the slope is very small, so this relationship is still statistically significant.

The slope of 0.6473 in the regression model means

for every 1 cm increase in a person's height, we expect their waist circumference to increase by approximately 0.65 cm, on average, based on the data.

Exercise 4: Conduct a hypothesis test to see if this relationship between wrist diameter and elbow diameter is statistically significant. Use wrist diameter as the independent (x) variable and elbow diameter as the dependent (y) variable.

Perform the full hypothesis test, including the following.

- Null and alternative hypotheses,
- point estimate for the slope $\hat{m}$ (the one we get from our linear model in R)
- standard error, test statistic, and degrees of freedom,
- p-value, and
- What is your conclusion about the relationship between wrists and elbows?

Null Hypothesis (H_0): $m = 0$

Alternative Hypothesis (H_1): $m \neq 0$

```

# Show full summary
> summary(wrist_elbow_model)

Call:
lm(formula = elb_di ~ wri_di, data = bdim)

Residuals:
    Min       1Q   Median       3Q      Max
-2.61361 -0.51032  0.02375  0.46111  2.42540

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.6993     0.3662    1.91  0.0567 .
wri_di        1.2033     0.0346   34.78 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7349 on 505 degrees of freedom
Multiple R-squared:  0.7055,    Adjusted R-squared:  0.7049
F-statistic: 1210 on 1 and 505 DF,  p-value: < 2.2e-16

```

Linear Regression Out

- Point estimate for slope (m): 1.2033
- Standard error of slope (SE): 0.0346
- Test statistic (t): 34.78
- Degrees of freedom (df): 505
- p-value: < 2e-16

Conclusion: Based on the linear regression output, there is strong statistical evidence of a positive linear relationship between wrist diameter and elbow diameter. The slope

estimate of 1.2033 indicates that for every additional centimeter in wrist diameter, the elbow diameter increases by approximately 1.2 cm on average.

Once you have concluded that a relationship exists, it is polite to follow up with a confidence interval for that slope. While the slope found by R is the best possible slope for *this* data, this data is a random sample from some larger data set, so our slope is just an estimate. The degrees of freedom for the t-distribution is $df = n - 2$. The confidence interval is determined as:

$$\text{point estimate } \hat{m} \pm (t_{crit}) \cdot SE$$

Exercise 5: Provide a 95% confidence interval for the slope from the previous exercise.

```
> summary(model)
```

Call:

```
lm(formula = elb_di ~ wri_di, data = bdims)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.61361	-0.51032	0.02375	0.46111	2.42540

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.6993	0.3662	1.91	0.0567 .
wri_di	1.2033	0.0346	34.78	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7349 on 505 degrees of freedom

Multiple R-squared: 0.7055, Adjusted R-squared: 0.7049

F-statistic: 1210 on 1 and 505 DF, p-value: < 2.2e-16

Slope: 1.2033

Standard Error: 0.0346

Degrees of Freedom: 505

Z = 1.96

We are 95% confident that for each additional centimeter of wrist diameter, the elbow diameter increases by between 1.14 cm and 1.27 cm in the population

